

EMILIO RANUCOLI

Machine Learning Engineer | Python Developer | Data & AI Specialist

Zaandam, Netherlands · Italian Citizen — EU Work Authorization (no sponsorship required)

Open to Remote EU & Hybrid · Trilingual: English · Spanish · Italian

+39 344 5721753 · ranucoliemilio@gmail.com · linkedin.com/in/emilio-ranucoli · github.com/RanuK12 · ranuk.dev

PROFESSIONAL SUMMARY

Machine Learning Engineer and Python Developer based in the Netherlands, with hands-on experience building production ML systems, ETL data pipelines and microservices for Booking.com, Accenture and enterprise clients across the EU. Specialized in Python, Scikit-learn, XGBoost, TensorFlow, FastAPI and AWS, with a strong record of measurable impact (12% lift in promotional targeting accuracy at Booking.com, 15% supply-chain cost reduction at Accenture, 2M+ daily transactions processed via near-real-time ETL). Founder of Ranuk IT Solutions, delivering custom software and ML products to clients in 4 countries. Italian citizen with full EU work authorization (no sponsorship required). Trilingual: English (fluent), Spanish & Italian (native).

"Combines technical rigor with an uncommon ability to understand the problem before writing code. Recommended 100%." — Delfina Bugliotti, Product Manager @ Booking.com

TECHNICAL SKILLS

Languages & ML: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, XGBoost, PyTorch), SQL, JavaScript / TypeScript, C#

Machine Learning: Supervised & Unsupervised Learning, Deep Learning, Neural Networks, NLP, Sentiment Analysis, Time Series Forecasting, MLOps

Data Engineering: ETL Pipelines, Data Wrangling, Data Modeling, Data Warehouse, RESTful APIs, JSON, Kafka, Streaming Data

Backend & Web: FastAPI, Flask, Node.js, React, Tailwind CSS, HTML5 / CSS3, Microservices, GraphQL

BI & Visualization: Power BI (PL-300 Certified), DAX, Tableau, Looker, Grafana, Streamlit, Plotly

Databases: PostgreSQL, MongoDB, MySQL, SQL Server, SAP

Cloud & DevOps: AWS (Cloud Practitioner Certified), Microsoft Azure (AI Fundamentals), GCP, Docker, Kubernetes, Git / GitHub, CI/CD

Testing & Methodologies: JUnit, TDD, REST, SOAP, Agile / Scrum, Code Reviews

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Contract) | Booking.com

Amsterdam, Netherlands · May 2025 – Oct 2025

- Led the development of a dynamic pricing optimization system, deploying Machine Learning models (Python, Scikit-learn, XGBoost, neural networks) for real-time price adjustments based on demand, competition and customer behavior.
- Analyzed 5M+ historical booking records to identify patterns that improved customer segmentation and promotional campaign effectiveness across multiple markets.
- Integrated production ML models into Booking.com's AWS environment with continuous monitoring, ensuring scalability and reproducibility of predictions.
- Built a GraphQL backend layer to reduce API latency and serve cached features to downstream services.
- Contributed to a 12% improvement in promotional targeting accuracy by mastering internal forecasting tools and tuning model features.

Founder & Lead Engineer | Ranuk IT Solutions

Barcelona, Spain (Remote) · 2024 – Present

- Founded and operate a tech consultancy delivering custom software and applied Machine Learning solutions to clients in Argentina, Spain, Italy and the United States.
- Delivered 28+ in-house projects across web, backend, automation and ML, owning the full lifecycle from product discovery to deployment and continuous support.
- NOT A ROBOT (international audiovisual production company): designed and built the full web platform powering their brand presence and client showcase.
- Bahay Design (architecture studio, Italy): delivered a high-end responsive website with same-day iteration cycles requested directly by the client.
- GARYCIO: built a backend service with WhatsApp bot integration, PostgreSQL persistence and automated PDF report generation for operational workflows.
- ranuk.dev: personal engineering brand featuring an interactive in-browser Neural Network Lab (configurable learning rate, epochs and hidden neurons with real-time decision-boundary visualization).
- Core stack: TypeScript, React, Node.js, Python, PostgreSQL, Scikit-learn, XGBoost, TensorFlow, PyTorch, MLOps.

Python Developer — Supply Chain Integration (Main role) | Accenture

Rome, Italy (On-site) · Nov 2024 – Apr 2025

- Designed and built Python microservices to integrate supply-chain data between SAP, Oracle and legacy ERPs for a major logistics client.
- Engineered automated ETL pipelines processing 2M+ daily transactions, reducing data latency from 24 hours to near real-time.
- Developed REST APIs for inventory synchronization across warehouses, achieving 99.7% data consistency across 50+ locations.
- Implemented Grafana dashboards and alerting systems for pipeline health monitoring and SLA compliance tracking.
- Stack: Python, FastAPI, SAP, Oracle, PostgreSQL, Docker, Grafana, REST APIs.

Python Developer (Freelance / Contract — in parallel) | Eurobrico S.p.A.

Villa Castel Ivano, Italy (Hybrid) · Nov 2024 – Apr 2025

- Delivered, in parallel with the Accenture Rome engagement, Python applications to optimize inventory management and retail operations for a major Italian DIY retail chain.
- Built data integration and automation solutions, reducing manual errors and improving operational efficiency across departments.
- Delivered the Eurobrico Web Redesign: UX audit and front-end prototype proposing a modernized customer journey.
- Collaborated directly with department heads to translate business needs into scalable technical solutions.

Data Science Analyst / Sr. Analyst — Supply Chain & Operations | Accenture

Argentina (Remote) · Jun 2024 – Nov 2024

- Developed predictive and optimization models for supply-chain operations using Python, SQL and Power BI.
- Implemented Machine Learning models for demand forecasting and inventory optimization, achieving a 15% operational cost reduction.
- Built interactive dashboards in Tableau and Power BI to visualize critical KPIs and support data-driven decision-making.
- Worked with multidisciplinary global teams under Agile / Scrum methodologies.

Data Analyst & Power BI Specialist | Government of Córdoba

Córdoba, Argentina · Nov 2023 – Jun 2024

- Designed and maintained 15+ Power BI dashboards tracking KPIs for public works and infrastructure programs.
- Performed data cleaning and modeling on 200K+ record datasets, improving report accuracy by 30%.
- Automated certification reporting workflows, reducing manual effort by 40% and freeing analyst time for higher-value work.

Purchasing & Inventory Analyst | Farmasut

Córdoba, Argentina · Nov 2015 – Oct 2023

- Conducted inventory audits and sales analysis with Tableau, reducing expiration losses by 15%.
- Optimized delivery routes through data analysis, decreasing logistics costs and improving fulfillment SLAs.

KEY PROJECTS

Crypto Analysis Dashboard · github.com/RanuK12

Stack: Python, Streamlit, Plotly, Binance API, Machine Learning, Telegram Bot

Cryptocurrency market analysis platform with real-time data ingestion, algorithmic trading strategy backtesting and a modern interactive web interface.

JobConnect — AI Job Finder · github.com/RanuK12

Stack: Flask, NLP, Web Scraping, Tailwind CSS

Web platform that uses NLP to match candidate CVs with job openings across multiple platforms, ranking results by semantic fit.

Movie Recommender · github.com/RanuK12/movie-recommender

Stack: Python, Scikit-learn, Pandas, Collaborative Filtering

ML-based recommendation engine producing personalized movie suggestions from user-rating patterns.

Sentiment Analysis Tool · github.com/RanuK12/review-sentiment-tool

Stack: Python, NLP, Scikit-learn

Customer-review classifier (positive / neutral / negative) usable as a microservice via REST API.

Laptop Cam Security · github.com/RanuK12/Laptop-Cam-Security

Stack: Python, OpenCV, Computer Vision, Automation

Local computer-vision tool that detects motion via webcam and triggers configurable security actions.

GARYCIO — WhatsApp Backend Bot · github.com/RanuK12

Stack: TypeScript, Node.js, WhatsApp API, PostgreSQL, PDF Generation

Backend service with WhatsApp integration and automated PDF reporting deployed for a production client of Ranuk IT.

EDUCATION

Systems Engineering — Universidad Tecnológica Nacional (UTN), Argentina

CERTIFICATIONS

- Microsoft Power BI Data Analyst (PL-300) — Microsoft (2024)
- AWS Cloud Practitioner — Amazon Web Services (2024)
- Microsoft Azure AI Fundamentals (AI-900) — Microsoft (2024)
- Machine Learning with Python — IBM (2025)
- Deep Learning Fundamentals — IBM (2025)
- Deep Learning with TensorFlow — IBM (2025)
- Intermediate Machine Learning — Kaggle (2024)

LANGUAGES

English: Fluent (professional working proficiency) **Spanish:** Native **Italian:** Native

PUBLICATIONS

"Y así voy tejiendo mi camino" — Memoir, self-published 2024. Available on Kindle.